

Lesson 3

Chapter 2 (continued): demand, Walras' law, and WARP

1 Demand functions

The consumer's choice depends on prices p and wealth (or income) w . We let $x(p, w)$ denote the *demand correspondence*: the set of bundles the consumer chooses from the budget set $B_{p,w}$. If the correspondence is single-valued — that is, the set consists of a single element — then we have a standard demand function. In the latter case

$$x(p, w) = (x_1(p, w), \dots, x_L(p, w)),$$

where $x_i(p, w)$ is the individual demand for good i , $i = 1, 2, \dots, L$.

We start with some basic assumptions, to the effect that the consumer is rational.

Homogeneity of degree 0

$x(p, w)$ is homogeneous of degree 0:

$$x(\alpha p, \alpha w) = x(p, w) \quad \text{for all } \alpha > 0.$$

This is saying that only relative prices and real income matter (the budget set is the same, $B_{\alpha p, \alpha w} = B_{p,w}$). It does not matter if we convert to a different currency, or from dollars to cents.

2 Walras' law and comparative statics

Walras' law

For any $p \gg 0$ and $w > 0$,

$$p \cdot x(p, w) = w,$$

i.e. the agent uses all his wealth (income).

We now assume that the demand correspondence is a demand function. We often assume it is differentiable.

Comparative statics

The parameters are (p, w) . What happens to the endogenous choices (i.e. consumption) when the exogenous parameters change? With differentiable demand functions we want to examine

$$\frac{\partial x_i(p, w)}{\partial p_j} \quad \text{and} \quad \frac{\partial x_i(p, w)}{\partial w}.$$

We now use the assumptions for comparative statics. Using homogeneity of degree 0,

$$x(\alpha p, \alpha w) = x(p, w) \quad \text{for } \alpha > 0 \quad \implies \quad \frac{dx_i(\alpha p, \alpha w)}{d\alpha} = 0,$$

where the derivative on the left is the *total* derivative with respect to α , i.e. the chain rule applied to all the arguments $\alpha p_1, \dots, \alpha p_L, \alpha w$. Hence

$$0 = \sum_{j=1}^L \frac{\partial x_i(\alpha p, \alpha w)}{\partial p_j} p_j + \frac{\partial x_i(\alpha p, \alpha w)}{\partial w} w.$$

Evaluate at $\alpha = 1$:

$$0 = \sum_{j=1}^L \frac{\partial x_i(p, w)}{\partial p_j} p_j + \frac{\partial x_i(p, w)}{\partial w} w.$$

When $x_i(p, w) > 0$, multiply the whole expression by $x_i(p, w)$ and divide each term inside by the same $x_i(p, w)$ — the factor cancels, so the equality is untouched:

$$\begin{aligned} 0 &= \sum_{j=1}^L \frac{\partial x_i(p, w)}{\partial p_j} p_j + \frac{\partial x_i(p, w)}{\partial w} w \\ &= x_i(p, w) \left[\sum_{j=1}^L \frac{\partial x_i(p, w)}{\partial p_j} \frac{p_j}{x_i(p, w)} + \frac{\partial x_i(p, w)}{\partial w} \frac{w}{x_i(p, w)} \right]. \end{aligned}$$

The bracket is the sum of the demand elasticities of good i with respect to every price (own and cross) and to wealth, and since $x_i(p, w) > 0$ it must itself be zero.

This is intuitive, given that the starting point is homogeneity of degree 0: nothing changes if all prices and wealth change by the same percentage. We learn that the elasticities cannot all have the same sign: unless they are all zero, some are positive and some negative.

3 The weak axiom of revealed preference (WARP)

WARP is a minimal rationality assumption: the agent does not make self-contradictory choices.

WARP is satisfied if, for any two pairs (p, w) and (p', w') , it holds that

$$p \cdot x(p', w') \leq w \quad \text{and} \quad x(p', w') \neq x(p, w) \quad \implies \quad p' \cdot x(p, w) > w'.$$

Tomorrow's bundle is affordable today, but I chose not to buy it: $x(p, w)$ is revealed preferred to $x(p', w')$. The reason I do not buy $x(p, w)$ tomorrow is simply that I cannot afford it.

4 WARP and Slutsky wealth compensation

WARP implies homogeneity of degree 0. If $(p', w') = (\alpha p, \alpha w)$, then x' is affordable whenever x was chosen, and vice versa. If $x' \neq x$, then x is revealed preferred to x' and x' is revealed preferred to x — a self-contradictory choice, which violates WARP. Hence $x' = x$.

WARP has strong implications when combined with Walras' law.

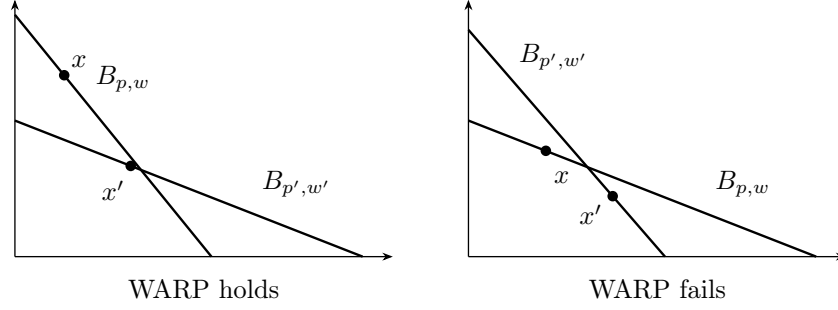


Figure 1: The two cases for a pair of price–wealth situations (p, w) and (p', w') . *Left:* x' is affordable when x is chosen under (p, w) , revealing that x is preferred to x' . Because x is unaffordable under (p', w') , there is no contradiction and WARP holds. *Right:* x' is affordable when x is chosen under (p, w) , and x is affordable when x' is chosen under (p', w') . This creates a mutual revealed preference, which is exactly what WARP forbids.

A thought experiment

Take a price change from p to p' and, at the same time, adjust income so that

$$w' = p' \cdot x(p, w),$$

which is exactly *Slutsky wealth compensation*: at the new prices the original bundle is just affordable. The result (the *compensated law of demand*): suppose $x(p, w)$ is homogeneous of degree 0 and satisfies Walras' law. Then $x(p, w)$ satisfies WARP if and only if, for any compensated price change from (p, w) to (p', w') , with $w' = p' \cdot x(p, w)$, it holds that

$$(p' - p) \cdot (x(p', w') - x(p, w)) \leq 0,$$

with strict inequality if $x(p, w) \neq x(p', w')$.

In particular, if only the price of good i changes ($p'_j = p_j$ for $j \neq i$), the inner product has a single term,

$$(p'_i - p_i)(x_i(p', w') - x_i(p, w)) \leq 0,$$

so the compensated demand for good i moves in the opposite direction to its own price. When several prices change at once only the inner product is signed, not each good separately.

Reading the picture

Because $p' \cdot x(p, w) = w'$, the original bundle lies on $B_{p', w'}$, so the two lines cross at $x(p, w)$. The new choice $x' = x(p', w')$ also lies on $B_{p', w'}$ (Walras' law). If $x' \neq x(p, w)$ were on the stretch inside $B_{p, w}$ (from the crossing up to the vertical intercept), each bundle would be affordable when the other was chosen, which WARP forbids. So x' lies on the thick stretch, where $p \cdot x' > w$. Here good 2 has become relatively more expensive (the new line is flatter), and the thick stretch has less of good 2 than $x(p, w)$: demand for the good whose price rose falls.

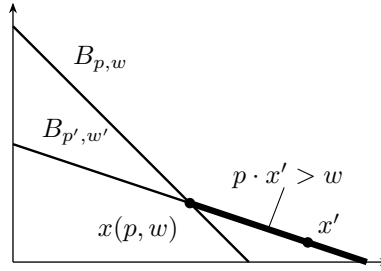


Figure 2: Slutsky wealth compensation. The new budget line $B_{p',w'}$ is drawn so that $w' = p' \cdot x(p, w)$: it passes through the original bundle $x(p, w)$, so the two lines cross exactly there. Write $x' = x(p', w')$ for the bundle chosen at the new prices. Since x' must satisfy $p' \cdot x' = w'$ it lies on $B_{p',w'}$, and since WARP requires x' to be unaffordable at (p, w) it can only lie on the thick part of that line, where $p \cdot x' > w$.

Transcription notes

- The original opens with a divider page reading only “Chapter 2”; it is represented here by the subtitle.
- “single-valuable” → *single-valued*; “weath” → *wealth*; “connect to different currency” → *convert*.
- “Walva’s” → *Walras*; “homogeneity” → *homogeneity*; “as verse versa” → *and vice versa*.
- “the agent uses all his wealth on income” → *uses all his wealth (income)*.
- The step after evaluating at $\alpha = 1$ is kept as in the original: multiply the whole expression by $x_i(p, w)$ and divide each term inside by the same $x_i(p, w)$. The two occurrences of the factor cancel, so the equality $= 0$ survives, and because $x_i(p, w) > 0$ the bracket — the sum of the elasticities — must itself be zero.
- **Corrected:** “all income elasticities sum to zero” is too narrow. What homogeneity gives is that the price elasticities of the demand for good i (own and cross) *plus* its income (wealth) elasticity sum to zero.
- **Corrected:** the compensated-change condition was written with “ $= 0$ ”. Since the next line says “strict if $x(p, w) \neq x(p', w')$ ”, the intended relation is $(p' - p) \cdot (x(p', w') - x(p, w)) \leq 0$: the equality sign was a slip.
- **Corrected:** the line “in the aggregate case” stated the sign condition good by good, $(p'_i - p_i)(x'_i - x_i) \leq 0$. That does not follow when several prices change together: only the inner product over all goods is signed. The good-by-good statement holds when only p_i changes, and it is written with that qualifier.
- **Corrected:** the statement of the compensated law of demand omitted homogeneity of degree 0, which the “if and only if” assumes (MWG Proposition 2.F.1); it is added. The sentence “for the compensated demand, assume that $x(p, w)$ satisfies Walras’ law” is folded into that statement.
- **Corrected:** “we learn that some elasticities are positive and some negative” is stated too strongly (they may all be zero); what follows from a zero sum is that they cannot all share a strict sign.
- “the set of bundles that are acceptable” is written as the set of bundles *chosen* from $B_{p,w}$; “real prices” is written *relative prices*. The derivative in α is typeset $d/d\alpha$, since it is a total derivative.
- The last sketch was redrawn with $x(p, w)$ at the crossing of the two budget lines — with Slutsky compensation the new line must pass through the original bundle — and with the stretch of $B_{p',w'}$ on which $x' = x(p', w')$ is admissible, i.e. where $p \cdot x' > w$, drawn thick and labelled, and x' shown on it.
- The summation indices are written inconsistently in the original (“ $\sum_{i=1}^L \partial x_i(\alpha p, \alpha w) / \partial p_i p_j$ ”, “ $\partial p_2 p_2$ ” after setting $\alpha = 1$, and “ $\sum_i \partial x_i / \partial p_i \cdot p_i / x_i$ ” in the elasticity line, which would leave only the own-price term); the wealth derivative is written “ $\partial x_i(\alpha p, w) / \partial w$ ”. All are typeset with the sum over j and the argument $(\alpha p, \alpha w)$.
- **Added:** the paragraph “Reading the picture” (why x' must lie on the thick stretch, and that the stretch has less of good 2) is an explanation of the final sketch, not written in the original.

- The WARP figure now follows the corrected reading. In the holds case only x' is affordable at (p, w) while x is unaffordable at (p', w') , so the antecedent of the axiom is met and its conclusion holds; in the failing case both bundles are affordable under both price-wealth pairs. The two panels use the same two lines with the labels swapped ($B_{p,w}$ the steeper line on the left, the flatter one on the right), so the two cases are read against the same geometry. The point labels are placed just clear of the lines. The panels are captioned “WARP holds” and “WARP fails” (the original labels only the second).